LECTURE NO 9: *Ethical Aspects of Technical Risks*

Engr. Tehseen Ahsan
Assistant Professor, Computer Engineering Department
HITEC University Taxila Cantt, Pakistan

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1.1 Introduction (1/3)

Case A Coal Mining

Between 1900 and 1975, about 600 million tons of coal were mined in the province of Limburg in the Netherlands (Pöttgens, 1988). In an area of about 220 km^2 the surface dropped about 2.5 m on average. In some locations there was a drop of more than 10 m. The coal mining operations were damaging the houses. Hundreds of millions in damages were paid by the companies mining the coal. The risk of damage to the houses was known in advance, but it thought to balance out against the benefits of the coal mining.

1.1 Introduction (2/3)

Case B DC-10 Disaster

On March 3, 1974 the freight door of a DC-10 opened during flight (Eddy, Potter, and Page, 1976). As a result, the plane crashed killing 346 people. The risk was known beforehand, at least to some of the people involved. On June 12, 1972 a similar accident had almost occurred. Already towards the end of the 1960s the possibility of this type of accident had been anticipated as a result of tests. Also the *Failure Mode and Effect Analysis* (FMEA) of the freight doors revealed the possibility of this type of accident. One important reason why nothing had been undertaken to reduce or avoid the risks was because there was an ongoing conflict between the supplier responsible for the door (Convair) and the plane manufacturer (Douglas). Neither party wanted to weaken its legal position. Neither therefore wanted to take the first step in the direction of adapting the design, even though each was aware of the shortcomings. The company's management ignored a memo written by a Convair engineer in which the shortcomings of the door design were outlined.

1.1 Introduction (3/3)

- These two cases demonstrate that to a certain extent hazards are inherent to technology. The two examples present hazards that were known beforehand. In **case A** those involved consciously took the risks of coal mining because of the expected advantages. In **case B**, with the DC-10, the risks were – in retrospect – regarded as unacceptable by most of the managers and engineers involved. However, the relevant risks were not removed or diminished for the reasons given in the example.
- Furthermore, less so than in a number of other cases, it is hard to ascribe the possible hazard to one specific technology. It is connected more with the large-scale use of a wide range of technologies.
- In this chapter, we discuss the moral issues that are raised by the risks and hazards of technologies, and how engineers can deal with those issues.

1.2 Definitions of Central Terms (1/2)

- ***Hazard-*** Possible damage or otherwise undesirable effect.
- ***Risk-*** The most often used definition of risk is the product of the probability of an undesirable event and the effect of that event.
- ***Safety-*** The condition that refers to a situation in which the risks have been reduced as far as reasonably feasible and desirable.
- ***Acceptable risk-*** A risk that is morally acceptable. The following considerations are relevant for deciding whether a risk is morally acceptable: *(1)* the degree of informed consent with the risk; *(2)* the degree to which the benefits of a risky activity weigh up against the disadvantages and risks; *(3)* the availability of alternatives with a lower risk; and *(4)* the degree to which risks and advantages are justly distributed.

1.2 Definitions of Central Terms (2/2)

- ***Uncertainty-*** A lack of knowledge. Refers to situations in which we know the type of consequences, but cannot meaningfully attribute probabilities to the occurrence of such consequences.
- ***Ignorance-*** Lack of knowledge. Refers to the situation in which we do not know what we do not know.
- ***Ambiguity-*** The property that different interpretations or meanings can be given to a term.

1.3 The Engineer's Responsibility for Safety (1/2)

- From where does the engineer's responsibility for safety come? Many codes of conduct for engineers attribute a responsibility for safety to engineers (see Lecture 4). Besides that, there are legal obligations concerning the safety of products or technical codes and standards, in which safety often plays an important role.
- Consequentialism states that engineers must strive for good consequences: safe products definitely fall into that category. The desirability to design safe products is sometimes described as "do no harm."
- Striving for safe products, therefore, is an important virtue.

1.3 The Engineer's Responsibility for Safety (2/2)

- During the design process, engineers can follow different strategies for ensuring safe products, such as:
  1. ***Inherently safe design:*** An approach to safe design that avoids hazards instead of coping with them, for example by replacing substances, mechanisms and reactions that are hazardous by less hazardous ones.
  2. ***Safety factors:*** constructions are usually made stronger than the load they probably have to bear. Adding a safety factor to the expected load or maximum load is an explicit way of doing this.
  3. ***Negative feedback mechanism-*** A mechanism that if a device fails or an operator loses control assures that the(dangerous) device shuts down.
  4. ***Multiple independent safety barriers-*** A chain of safety barriers that operate independently of each other so that if one fails the others do not necessarily also fail.

1.4 Risk Assessment (1/7)

- ***Risk assessment-*** A systematic investigation in which the risks of a technology of an activity are mapped and expressed quantitatively in a certain risk measure.
- A risk assessment usually consists of four steps:
  1. Release assessment
  2. Exposure assessment
  3. Consequence assessment
  4. Risk estimation

1.4 Risk Assessment (2/7)

1. Release Assessment

- Releases are any physical effects that can lead to harm and that originate in a technical installation. Examples are shock waves, radiation, and the spread of hazardous substances. In general, we can distinguish between two kinds of releases: ***incidental*** and **continuous**. *Incidental releases are usually unintended and are due to, for example, an explosion in a chemical plant or an accident with a nuclear power plant. Such releases can often cause immediate and major harm*. *Continuous releases are often anticipated and may be accepted as side-effects of, for example, production processes. Continuous releases do not necessarily or always lead to exposure or harm*.
- In the case of incidental releases, an important step is the detection of so called failure modes and accident scenarios.

1.4 Risk Assessment (3/7)

1. Release Assessment (continued…)

- The probability of the occurrence of such scenarios is calculated too. This takes place in two ways. In the first method, the probability of certain accidents occurring is calculated using statistical data about accidents in the past. If such statistical data are absent *event trees* and *fault trees* are often used to calculate the probability of an accident.
- ***Event tree-*** Tree of events in which one starts with a certain event and considers what events will follow.
- ***Fault tree-*** Tree of events in which we move backwards from an unwanted event (a fault) to the events that could lead to the undesirable event.

1.4 Risk Assessment (4/7)

2. Exposure Assessment

- In this step the aim is to predict the exposure of vulnerable subjects like human beings to certain releases. Exposure assessment usually describes what vulnerable subjects(human beings, animals, the environment) are exposed to a certain release, through what mechanisms (for example, inhalation of toxic substances by humans), and the intensity, frequency, and duration of the exposure.

1.4 Risk Assessment (5/7)

3. Consequence Assessment

- In the third step the focus is on determining the relationship between exposure and harmful consequences. In some risk assessments, this assessment is limited to acute harm or to the number of direct fatalities. In other cases, long-term effects on health or the environment are also considered. An important part of this step is usually the determination of dose-response relationships. Such relationships can be established through tests on animals, epidemiology and models.

1.4 Risk Assessment (6/7)

4. Risk Estimation

- In the fourth step the risk is determined and presented using the results obtained earlier. In this step we determine in what measure the risk is expressed. This can be done using the number of expected fatalities per time unit, for example, or the reduced lifespan of people that work or live in the neighbourhood of an installation.
- In many cases, risk assessments only have limited reliability. This is because the results often depend on the original assumptions made.
- Many risk assessments do not give an estimate of the accuracy and reliability of the final result.

1.4 Risk Assessment (7/7)

4. ***Risk Estimation (continued…)***

- In establishing a risk on the basis of a body of empirical data one might make two kinds of mistakes.
  1. ***Type-I Error-*** one can establish a risk when there is actually no risk.
  2. ***Type-II Error-*** one can mistakenly conclude that there is no risk while there actually is a risk.

1.5 When are Risks Acceptable? (1/7)

- Some engineers and scientists believe that if one activity is acceptable the other (which has the same risk) must be acceptable too. This argument is flawed for a number of reasons. First, the question whether the risks of technology A are acceptable is not the same question as the question whether technology A is acceptable. This will become clear when we consider the ethical objections to human cloning.

1.5 When are Risks Acceptable? (2/7)

Ethical Objections to Cloning

- Philosopher of technology Tsjalling Swierstra has made the following list of arguments:
  1. undermines the uniqueness of humans;
  2. is contrary to human dignity;
  3. leads to psychological problems in the cloned child;
  4. suffers from numerous scientific and technical risks;
  5. will lead to misuse and has unforeseen and undesirable consequences;
  6. is unnatural;
  7. is based on an overreaching desire for manipulation and leads to the abject instrumentalization of people;
  8. will lead to undesirable changes in our self-image.

1.5 When are Risks Acceptable? (3/7)

1.5.1 Informed Consent

- Principle that states that activities (experiments, risks) are acceptable if people have freely consented to them after being fully informed about the (potential) risks and benefits of these activities (experiments, risks).
- There are different ideas about how the principle of informed consent should be applied in technology. One idea is to allow this to occur through the economic market. In such a scenario, it is assumed people will decide for themselves which risks they wish to take concerning the purchase of certain risky or dangerous products.

1.5 When are Risks Acceptable? (4/7)

1.5.1 Informed Consent (continued…)

- First, consumers are often insufficiently or incompletely aware of the risks of technical products. So we cannot speak of real consent with the risks involved. People, for example, buy cell phones without being aware of any possible health risks. In such a case we cannot say that people have consented to those health risks. Second, technical products often introduce risks that affect people other than the buyers or sellers of the products in question without their consent. One example is paint with organic solvents that contribute to environmental problems such as smog.

1.5 When are Risks Acceptable? (5/7)

1.5.2 Do the advantages outweigh the risks?

- An important reason why risks can be ethically acceptable is that risky activities can have advantages. More generally we could argue that risky activities and thus the risks that are linked to these activities are acceptable if the benefits of the activities outweigh the costs.
- Instruments have been developed to determine the most desirable level of risk for a technology. One example of this is *risk-cost-benefit analysis*.
- ***Risk-cost-benefit analysis-*** The social costs for risk reduction are weighed against the social benefits offered by risk reduction, so achieving an optimal level of risk in which the social benefits are highest.

1.5 When are Risks Acceptable? (6/7)

1.5.3 The availability of alternatives

- The acceptability of the risks of a technology also depends on the availability of alternatives with lower risks. Suppose that two technologies, say A and B, introduce the same risks.
- Now also suppose that for technology A an alternative is available with lower risk and no major other disadvantages, while for technology B such an alternative is absent. In this case, we may well conclude that the risks of technology A are unacceptable (because an alternative with lower risks is available) while the risks of technology B are acceptable (since no alternative is available and the technology is in other respects acceptable).

1.5 When are Risks Acceptable? (7/7)

1.5.4 Are risks and benefits justly distributed?

- An important ethical consideration in accepting risks is the degree to which risks and benefits of risky activities are justly distributed. It is, for example, not just if certain groups of people always have to carry the load of certain activities, while other groups reap the benefits.
- People do not get to choose which risks they find acceptable – the regulator, often being the government, does that for them instead. This objection is especially applicable to ***personal risks-*** Risks that only affect an individual and not a collective. For example, the risk of smoking.
- ***Collective Risks-*** Risks that affect a collective of people and not just individuals, like the risks of flooding.

1.6 Risk Communication

- According to some professional codes of conduct, engineers must inform the public about risks and hazards. In some cases specialists are used, who are called *risk communicators*.
- ***Risk Communicator-*** specialists that inform the public about risks and hazards.
- Risk communication must first be honest (do not lie and always tell the complete truth). Next to that, it must respect the freedom of choice and autonomy of people and hence not be paternalistic. Here, the principle of informed consent is of importance too. It implies that you must not try to convince people but only inform them.
- Attempt to convince people by means of risk communication or withholding certain information can be morally right if it results in good consequences.

1.7 Dealing with Uncertainty and Ignorance (1/4)

- Up to now, we have presupposed that it is possible to predict and express risks related to the technologies to a certain extent. But what happens if that is not the case? Consider the supposed health issues surrounding cell phones and the possible negative health and environment effects of growing and consuming genetically manipulated crops. In some of these cases, risks are calculated, but the question is whether all possible hazards have been assessed in a reliable way.

1.7 Dealing with Uncertainty and Ignorance (2/4)

1.7.1 The precautionary principle

- Principle that prescribes how to deal with threats that are uncertain and/or cannot be scientifically established. In it's most general form the precautionary principle has the following general format:
- If there is *(1)* a threat, which is *(2)* uncertain , then *(3)* some kind of action is mandatory. This definition has four dimensions: *(1)* the threat dimension; *(2)* the uncertainty dimension; *(3)* the action dimension; and *(4)* the prescription dimension.

1.7 Dealing with Uncertainty and Ignorance (3/4)

1.7.2 Engineering as a societal experiment

- Like science, engineering has an experimental nature. In science, hypotheses are deduced from theories, which are then tested in experiments. A hypothesis can be confirmed or falsified through the experiment. Analogous to this, newly designed products can be seen as hypotheses for properly functioning products. The hypothesis that they are properly functioning is usually tested through simulations and experiments in the laboratory, in small-scale field tests.
- Although such simulations, experiments, tests, and trials are very important in engineering, they do not always provide complete and reliable knowledge of the functioning of technological products and the hazards and risks involved.

1.7 Dealing with Uncertainty and Ignorance (4/4)

1.7.2 Engineering as a societal experiment (continued…)

- For a variety of reasons, it is often not possible to completely predict the possible hazards of new technologies before they enter society.
- In some cases society has become the laboratory for engineering experiments. This means that it is ultimately during actual implementation of a technology in society that its functioning and possible hazards and risks are tested. However, unlike traditional scientific experiments, societal experiments are usually difficult to terminate if something goes wrong. Moreover, in societal experiments with new technologies, the consequences can be much larger and can have an impact on third parties.
- Societal experiments in technology are nearly always large scale, can have irreversible negative consequences, and usually involve people as experimental subjects.

References: Textbooks

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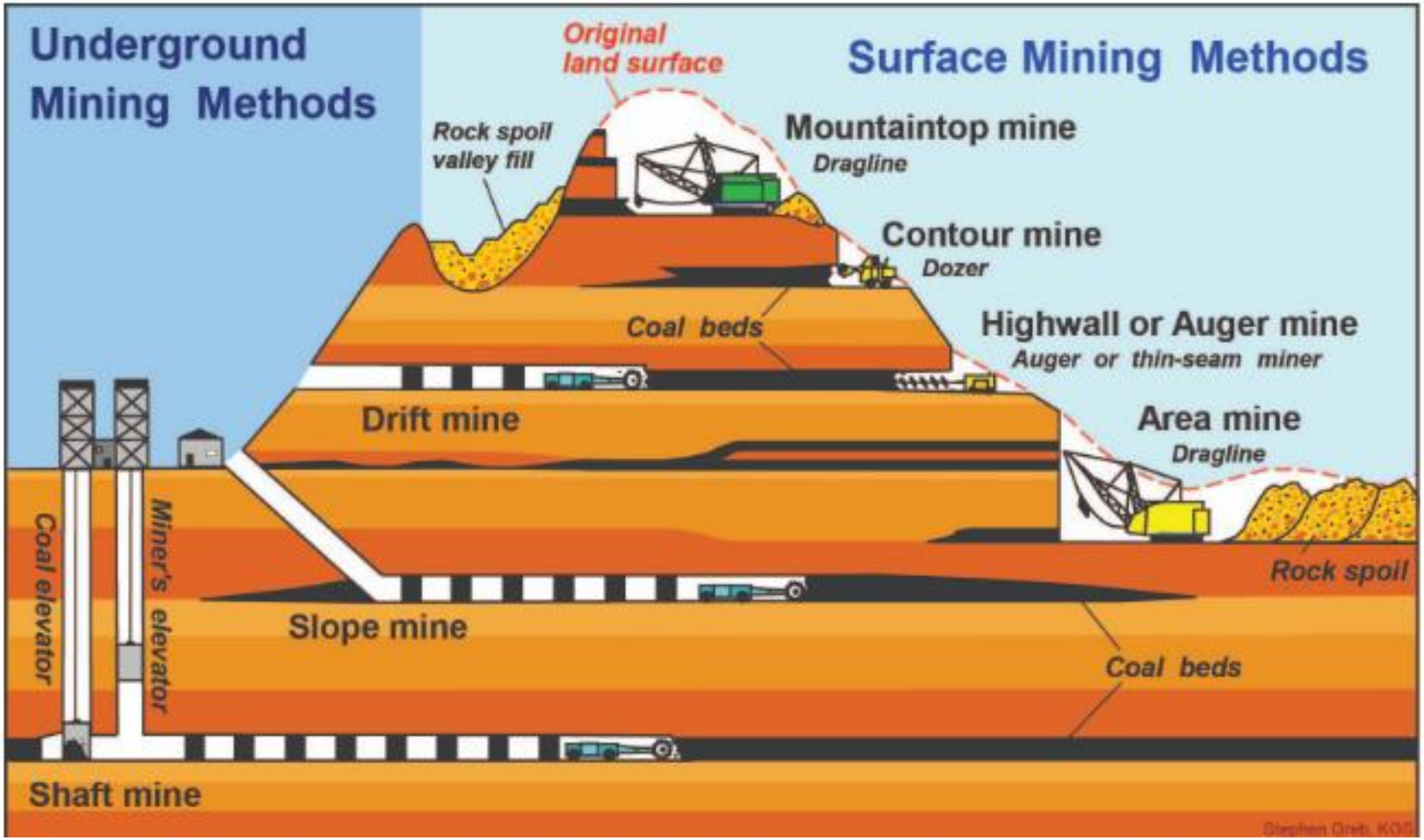
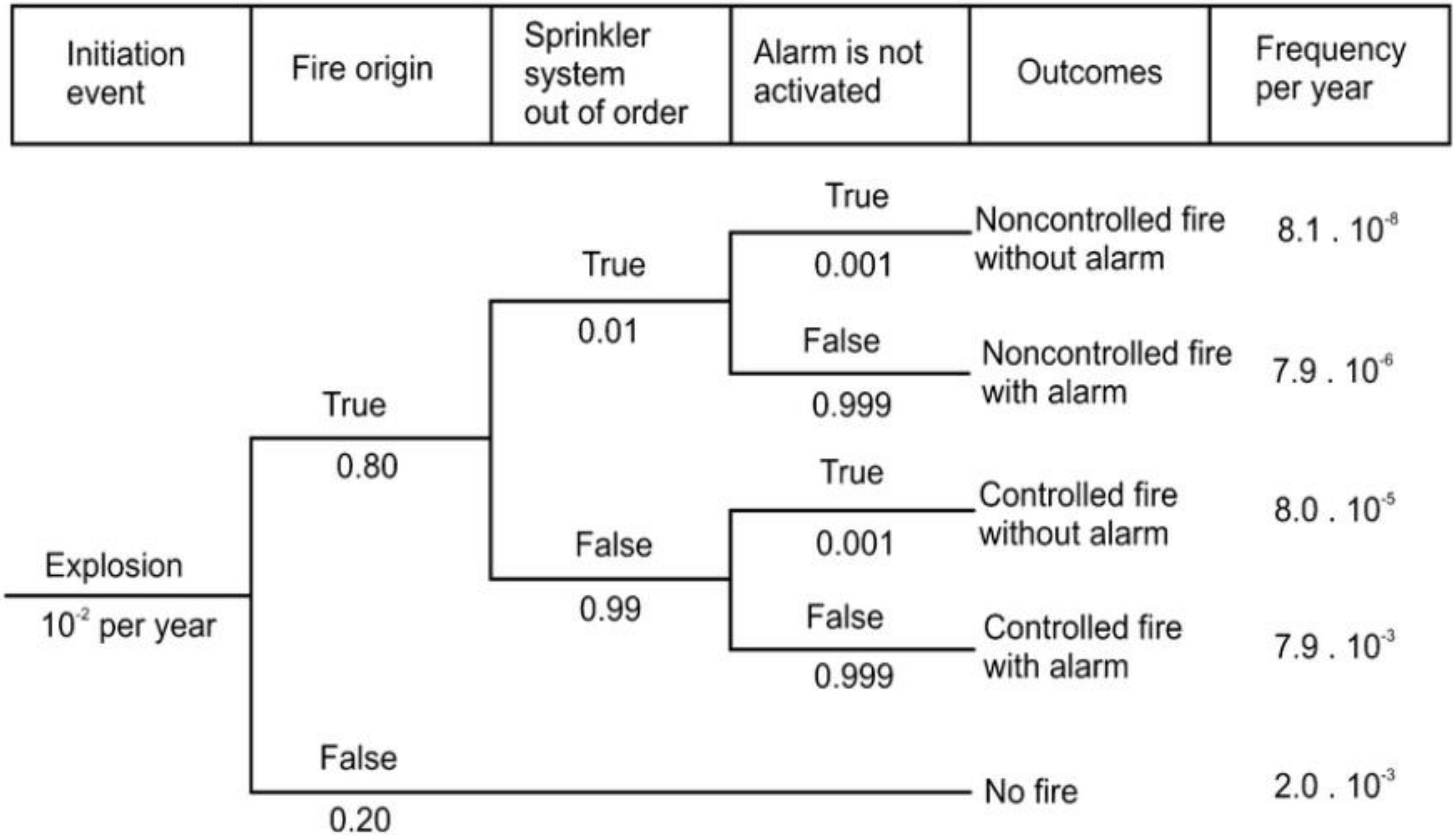
APPENDIX

How coal mining operations resulted in dropped surfaces?



Example of an event tree analysis of an explosion



Example of a fault tree analysis